

TechBiz Security
Cyber Security Internship Program

Week 3 – Task 9 (EH)
Enumeration — Digging Deeper into Services

Practical Report

Field	Details
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Internship ID	TBS-INT-2026-518
Submitted Date	07 September 2026

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1. Introduction

Service enumeration is the process of actively interrogating open ports and running services on a target system to extract detailed, actionable information about them. While a port scan identifies which ports are open, enumeration goes a step further by attempting to determine exactly what is running behind each port and what additional information can be gathered from it.

During enumeration, a tester typically attempts to identify the following types of information:

- Service names and running processes
- Exact service versions
- Server banners returned by the service
- SMB workgroup or domain information
- SMB shares available on the host
- Resources that can be accessed, including anonymously
- Potential usernames associated with the target
- Other service-specific details that may assist further assessment

This information is critical during a penetration test because it allows the tester to build an accurate picture of the target's attack surface. Knowing the exact software version running on a service, for example, allows a tester to check for known vulnerabilities associated with that version. Similarly, discovering accessible SMB shares or valid usernames can open up further avenues for exploitation or lateral movement. This task focused on practising enumeration techniques against a deliberately vulnerable lab machine in a safe, authorised environment.

2. Task Objectives

- Explore open services discovered during scanning.
- Perform service-version identification using Nmap.
- Perform SMB enumeration on the target host.
- Use enum4linux to gather SMB information such as workgroup, shares, and usernames.
- Use smbclient to enumerate and access available SMB shares.
- Perform FTP banner grabbing using Netcat.
- Identify useful information such as versions, workgroups, shares, and usernames.
- Practice enumeration concepts using the TryHackMe Vulniversity and Nmap rooms.

3. Lab / Practical Environment

The practical exercises in this task were carried out against a deliberately vulnerable lab machine, used strictly for authorised cybersecurity training purposes. The environment details are summarised below.

Parameter	Details
Attacker Machine	Kali Linux
Target Machine	Metasploitable 2

4. Initial Service Enumeration with Nmap

```
sudo nmap -sV -p- 10.104.63.43
```

```
manu@kali:~$ sudo nmap -v -p- 10.104.63.43
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-07 20:35 +0300
Nmap scan report for 10.104.63.43
Host is up (0.0039s latency).
Not shown: 65535 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ftp
22/tcp    open  ssh
22/tcp    open  telnet
23/tcp    open  smtp
33/tcp    open  domain
80/tcp    open  httpd
117/tcp   open  rackdb
139/tcp   open  netbios-ssn
145/tcp   open  netbios-ssn
312/tcp   open  exec
513/tcp   open  login
518/tcp   open  tcpwrapped
8099/tcp   open  java-rmi
1224/tcp  open  bindshell
2049/tcp  open  nfs
2121/tcp  open  ftp
2306/tcp  open  mysql
3032/tcp  open  distccd
3432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  x11
6067/tcp  open  irc
6067/tcp  open  irc
6069/tcp  open  ajp13
8108/tcp  open  http
8787/tcp  open  drb
8529/tcp  open  lockmgr
38231/tcp open  status
4040/tcp  open  java-rmi
71719/tcp open  mountd
MAC Address: 08:00:27:02:14:10 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 143.74 seconds

manu@kali:~$
```

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5. SMB Enumeration using enum4linux

enum4linux is a tool used specifically to enumerate information from Windows and Samba (SMB) hosts, including workgroup/domain details, NetBIOS information, share names, users, and password policies. It was run against the target with the -a flag, which performs all standard enumeration checks in a single run.

```
enum4linux -a 10.104.63.43
```

The output confirmed several important pieces of information about the target:

- The SMB workgroup was identified as WORKGROUP.
- NetBIOS name resolution identified the host as METASPLOITABLE, with services such as Workstation Service, Messenger Service, File Server Service, and Master Browser roles active.
- A null/anonymous session was permitted on the target using an empty username and password, which is a significant misconfiguration.
- A list of known/common usernames (administrator, guest, krbtgt, domain admins, root, bin, none) was returned by the RID range enumeration.
- The domain SID could not be determined, and the tool could not conclusively establish whether the host was part of a domain or only a workgroup.

In parallel, smbclient was used with the -L flag to directly list the shares exposed by the target over SMB, and Netcat was used to grab the FTP banner on port 21. Both outputs are shown together in the screenshot below.

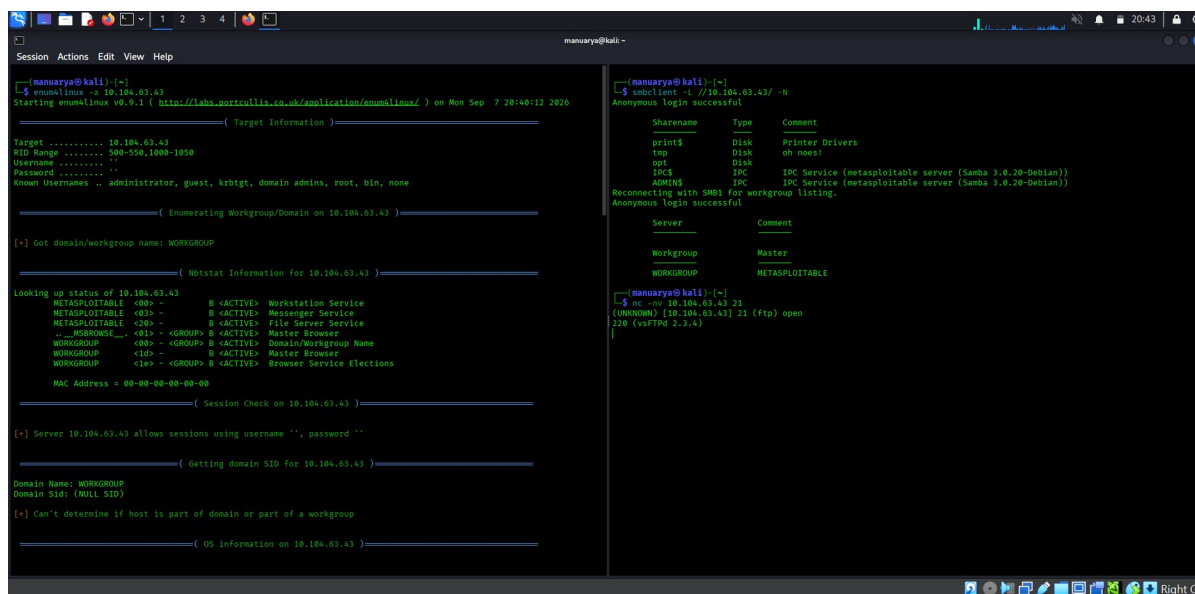


Figure 2: (Left) enum4linux output showing workgroup, NetBIOS, and username information. (Right) smbclient share listing (print\$, tmp, opt, IPC\$, ADMIN\$) and Netcat FTP banner grab confirming vsFTPd 2.3.4.

The smbclient listing (right-hand pane, top) confirmed anonymous login was successful and enumerated five shares: print\$, tmp, opt, IPC\$, and ADMIN\$, along with the comment on ADMIN\$ and IPC\$ identifying the host as a Samba 3.0.20-Debian server. It also confirmed the workgroup as WORKGROUP with METASPLOITABLE as the master browser. The Netcat command below it connected directly to port 21 and captured the raw FTP banner, confirming vsFTPd 2.3.4 — matching the version identified earlier by Nmap.

6. SMB Share Enumeration using smbclient

After identifying the available shares, the tmp share was accessed directly using smbclient with anonymous (null session) authentication, using the -N flag to suppress the password prompt:

```
smbclient //10.104.63.43/tmp -N
```

The connection succeeded without requiring a valid username or password, confirming that anonymous access to this share was permitted. Once connected, the ls command was issued inside the SMB session to list the contents of the share.

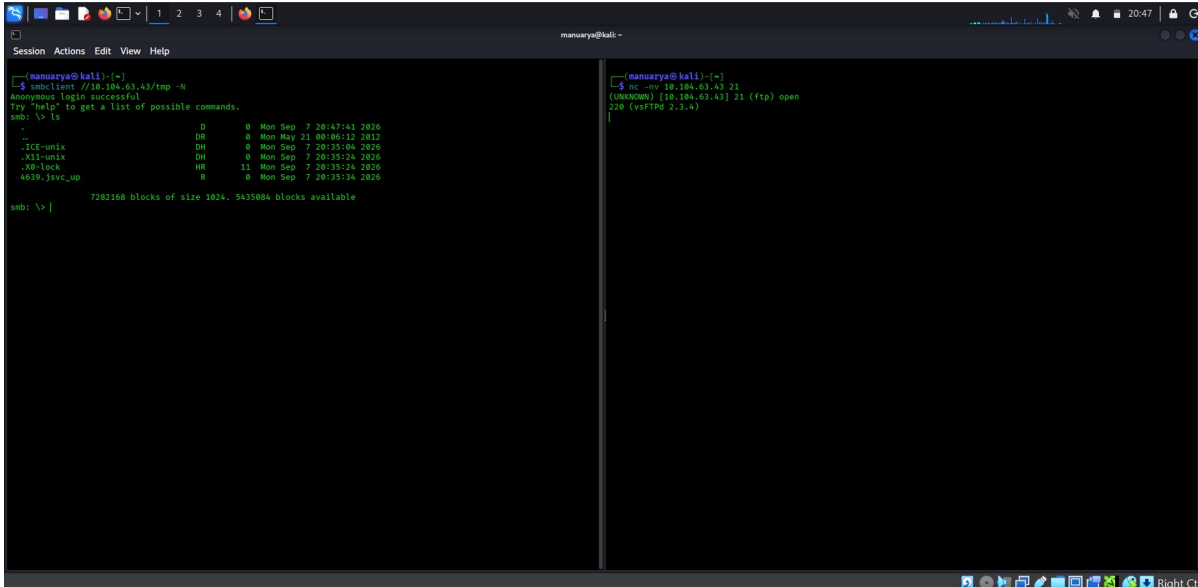


Figure 3: (Left) Anonymous smbclient session on the tmp share, with directory listing showing hidden lock/socket files. (Right) Netcat session confirming the vsFTPd 2.3.4 FTP banner on port 21.

The directory listing returned several system-related entries, including .ICE-unix, .X11-unix, and .X0-lock, along with available disk space information. This confirms that the tmp share was both browsable and readable without authentication, which represents a genuine security weakness that would be flagged in a real-world penetration test report. This concludes the SMB enumeration phase, where anonymous access, share names, workgroup details, and potential usernames were all successfully identified.

7. FTP Banner Grabbing using Netcat

Banner grabbing is a technique used to identify the exact software and version running behind a service by connecting to it directly and reading the initial response (banner) it sends. Netcat was used to connect to port 21 on the target:

```
nc -nv 10.104.63.43 21
```

The connection returned the banner 220 (vsFTPd 2.3.4), confirming the FTP service and its exact version. This result is consistent across both Figure 2 and Figure 3, and matches the version identified independently by the Nmap service scan in Section 4, providing cross-validation of the finding using two different techniques.

8. TryHackMe Practice — Vulniversity and Nmap Rooms

Alongside the manual lab exercises, the concepts covered in this task were reinforced using two rooms on the TryHackMe platform.

Vulniversity Room

The Vulniversity room covers active reconnaissance, web application attacks, and privilege escalation, and includes practical enumeration exercises similar to those performed in this task. As shown in the screenshot below, this room was completed to 100%, with all listed tasks (Deploy the machine, Reconnaissance, Locating directories using Gobuster, Compromise the Webserver, and subsequent tasks) marked as finished.

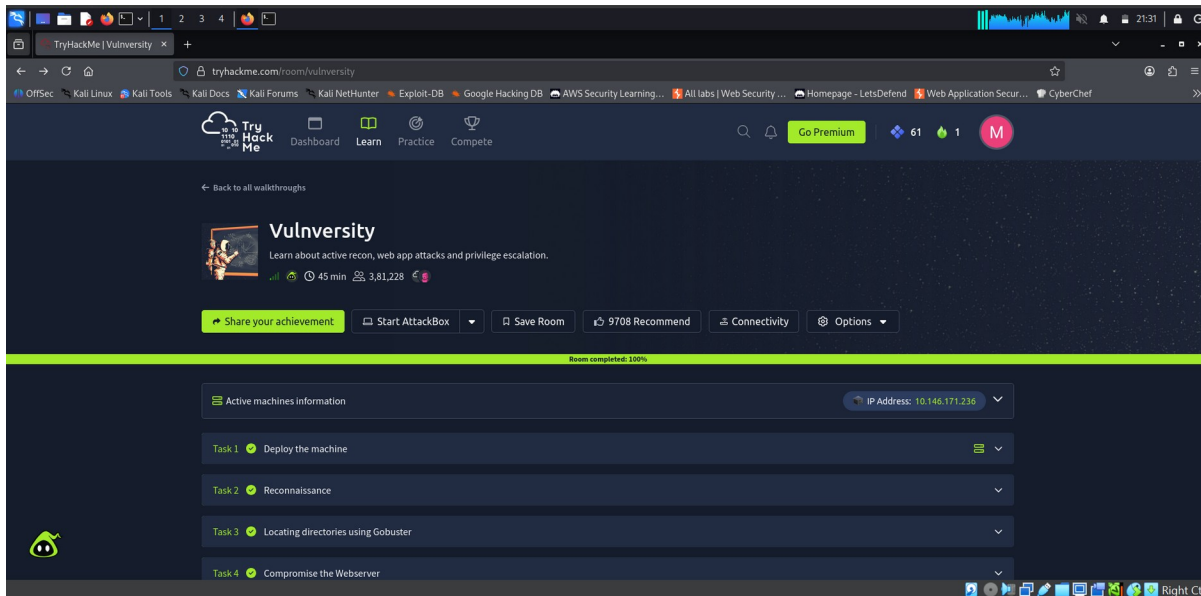


Figure 4: TryHackMe Vulniversity room showing Room completed: 100%, with all tasks marked complete.

Nmap Room

The Nmap room provides an in-depth look at scanning with Nmap, covering switches, scan types, and service/version detection — directly supporting the scanning work carried out in Section 4 of this report. As shown in the screenshot below, progress in this room currently stands at 95%, with the Deploy, Introduction, and Nmap Switches tasks completed and the Scan Types task in progress.

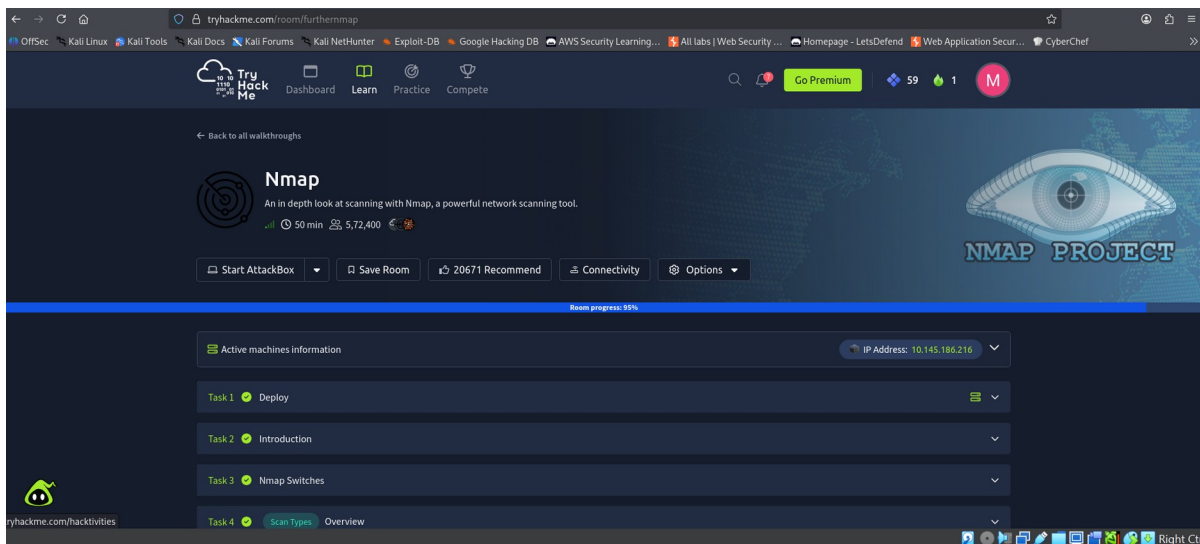


Figure 5: TryHackMe Nmap room showing Room progress: 95%, with the Scan Types task currently in progress.

9. Key Findings Summary

Finding	Detail
FTP Service	vsFTPD 2.3.4 (confirmed by both Nmap and Netcat banner grab)
SSH Service	OpenSSH 4.7p1 on Debian
SMB Workgroup	WORKGROUP (NetBIOS name: METASPLOITABLE)
SMB Shares Identified	print\$, tmp, opt, IPC\$, ADMIN\$
Anonymous SMB Access	Permitted — null session allowed, tmp share readable without credentials
Possible Usernames	administrator, guest, krbtgt, domain admins, root, bin, none
Samba Version	Samba 3.0.20-Debian (from share comments)

10. Conclusion

This task provided practical, hands-on experience in service enumeration beyond basic port scanning. Using Nmap, the exact service and version running on each open port were identified. Using enum4linux and smbclient, detailed SMB-specific information — including workgroup name, NetBIOS details, share names, and possible usernames — was gathered, and it was confirmed that anonymous access to certain SMB shares was permitted. Finally, Netcat was used to manually grab the FTP banner, cross-confirming the FTP service version found earlier through Nmap.

Overall, this exercise reinforced the understanding that enumeration is a multi-tool process, and that combining several techniques (automated scanning, dedicated SMB enumeration tools, and manual banner grabbing) provides a more complete and reliable picture of a target's attack surface than any single tool alone. The misconfigurations identified — particularly anonymous SMB access — are exactly the type of finding that would be documented and flagged in a real penetration testing engagement.